

Medvale Internal Medicine - Chapter 11E

Pigment, Hair, Nail, Benign Lesions, and Skin Cancer

Part XI Dermatology | Original synthesis for study and Graph-RAG extraction

This chapter collects dermatology topics that often appear as image recognition questions but require a structured internal medicine approach: pigmentary change, alopecia, nail disease, benign growths, premalignant lesions, nonmelanoma skin cancer, melanoma, and cutaneous malignancy mimics. The practical goal is to describe lesions precisely, identify dangerous patterns that require urgent biopsy or referral, and connect morphology to systemic associations. The chapter continues the Medvale style by emphasizing narrative reasoning, comparison tables, and diagnostic algorithms rather than isolated visual memorization.

Source integration note: The uploaded dermatology source for the preceding segment emphasized warts, molluscum, zoster, dermatophytes, tinea versicolor, scabies, actinic keratosis, seborrheic keratosis, and keratoacanthoma. This chapter extends that lesion framework into pigment, hair, nail, and malignant dermatology while preserving the same high-yield exam distinctions.

1. General Diagnostic Approach to Localized Lesions and Pigment Change

A dermatologic diagnosis begins with morphology and distribution before naming a disease. The same brown papule can be a benign nevus, seborrheic keratosis, pigmented basal cell carcinoma, melanoma, dermatofibroma, or post-inflammatory change. The same white patch can be vitiligo, tinea versicolor, post-inflammatory hypopigmentation, pityriasis alba, chemical leukoderma, or hypopigmented mycosis fungoides. The first step is therefore to describe whether the process is flat or raised, inflammatory or noninflammatory, melanocytic or nonmelanocytic, solitary or generalized, stable or evolving, symptomatic or asymptomatic, and whether it is located in a high-risk site such as the face, ear, lip, genital skin, palm, sole, nail unit, or chronically scarred/inflamed skin.

For Graph-RAG purposes, each lesion can be indexed along several axes: morphology, color, distribution, tempo, symptoms, risk factors, histology/biopsy requirement, and treatment. For example, melanoma is a malignant melanocytic neoplasm characterized clinically by asymmetry, border irregularity, color variegation, diameter enlargement, and evolution; histologically it is staged by Breslow thickness; management begins with excisional biopsy and definitive wide local excision. In contrast, seborrheic keratosis is a benign keratinocytic proliferation with a stuck-on waxy/verrucous surface and no malignant potential, although irritated or atypical lesions may still be biopsied if diagnosis is uncertain.

First question	What to ask or examine	High-yield implications
Is the lesion flat or raised?	Macule/patch versus papule/plaque/nodule/tumor.	Flat pigment change often suggests melanocytic, vascular, post-inflammatory, infectious, or drug-related etiologies; raised lesions require assessment of surface texture and depth.
Is pigment uniform?	Single color versus variegated black, brown, blue, red, white, or gray.	Multiple colors and regression-like white/blue-gray areas raise concern for melanoma.
Is it changing?	Growth, bleeding, ulceration, new pain, itch, crusting, or failure to heal.	Evolution is one of the strongest warning signs for malignancy and often overrides reassuring symmetry.
Where is it?	Sun-exposed skin, palms/soles, nails, mucosa, scars, genital region.	Location narrows differential: acral/nail lesions may represent acral lentiginous melanoma; chronic sun-exposed areas favor AK, BCC, SCC.
Is there systemic context?	Immunosuppression, HIV, transplant, autoimmune disease, pregnancy, endocrine disease, drug exposure.	Immunosuppression increases SCC, Kaposi sarcoma, HPV-related lesions, zoster, molluscum, and aggressive skin cancers.

2. Pigmentary Disorders

Pigmentary disorders are easiest to separate into hypopigmented/depigmented disorders and hyperpigmented disorders. Hypopigmentation means less pigment than surrounding skin but not complete absence; depigmentation implies near-complete loss of melanin, often accentuated by Wood lamp. Hyperpigmentation may reflect increased melanin, hemosiderin, exogenous pigment, drug deposition, or vascular/purpuric change that appears dark rather than truly melanocytic. Distribution and borders are often more useful than color alone.

Condition	Classic morphology/distribution	Pathophysiology and associations	Diagnosis and management
Vitiligo	Well-demarcated depigmented macules or patches; often periorificial, acral, extensor, genital, or sites of trauma; hairs may turn white.	Autoimmune melanocyte destruction; associated with autoimmune thyroid disease, type 1 diabetes, pernicious anemia, alopecia areata, and other autoimmune disorders.	Clinical diagnosis aided by Wood lamp accentuation. Treat with sun protection, topical corticosteroids or calcineurin inhibitors, phototherapy for extensive disease, topical ruxolitinib for selected nonsegmental disease, and screening guided by symptoms/risk.
Tinea versicolor	Hypo- or hyperpigmented finely scaling macules/patches on trunk, neck, shoulders; more visible after sun exposure.	Malassezia yeast overgrowth; humid environments, sweating, oily skin. Pigment change can persist after organism clearance.	KOH shows short hyphae and spores. Treat with topical selenium sulfide, ketoconazole, or azoles; oral therapy for extensive/recurrent cases.
Post-inflammatory pigment alteration	Dark or light macules after acne, eczema, psoriasis, trauma, burns, or procedures.	Melanin increase or decrease after inflammation; more prominent and persistent in darker skin.	Treat underlying inflammation, photoprotection, time. Hyperpigmentation may respond to topical retinoids, azelaic acid, hydroquinone, or procedures with caution.
Melasma	Symmetric reticulated or confluent hyperpigmented patches on cheeks, forehead, upper lip, chin; sharply worsens with light exposure.	Hormonal and ultraviolet/visible light triggers; pregnancy and estrogen exposure are common associations.	Strict broad-spectrum/tinted photoprotection. Topicals include hydroquinone-based combinations, azelaic acid, retinoids; recurrence is common.

Condition	Classic morphology/distribution	Pathophysiology and associations	Diagnosis and management
Acanthosis nigricans	Velvety hyperpigmented plaques in neck, axillae, groin, inframammary areas.	Insulin resistance most common; also obesity, PCOS, endocrinopathies, medications. Abrupt extensive disease can be paraneoplastic, especially gastric adenocarcinoma.	Screen for metabolic syndrome/diabetes when appropriate. Treat underlying insulin resistance; biopsy or malignancy evaluation if sudden, severe, mucosal, or unexplained.
Cafe-au-lait macules	Uniform tan-brown macules/patches, smooth borders.	Multiple lesions may suggest neurofibromatosis type 1 or other genetic syndromes depending on number, size, axillary freckling, neurofibromas, or family history.	Clinical recognition; evaluate syndromic criteria when numerous or accompanied by neurologic/ocular findings.

Pitfall: A depigmented or hypopigmented lesion that is persistent, scaly, and located in a bathing-trunk distribution can rarely represent hypopigmented mycosis fungoides rather than eczema or tinea. Chronic, treatment-resistant patches deserve biopsy.

Pigment algorithm: white, brown, black, blue, and red lesions

If a lesion is white, determine whether there is scale. Scale supports tinea versicolor, eczema, psoriasis, or pityriasis alba; no scale with sharp borders supports vitiligo. If a lesion is brown or black, determine whether it is melanocytic, keratinocytic, vascular, or foreign pigment. Melanocytic lesions are assessed using ABCDE, ugly duckling comparison, dermoscopy if available, and biopsy when suspicious. Blue-black nodules may represent blue nevus, nodular melanoma, thrombosed angioma, pigmented BCC, or traumatic tattoo. Red-purple nonblanching lesions suggest purpura, vasculitis, Kaposi sarcoma, angioma, or hematoma rather than melanin.

3. Alopecia and Hair Disorders

Hair loss should be divided into scarring versus nonscarring alopecia. Nonscarring alopecia preserves follicular openings and is potentially reversible because the follicular stem cell niche remains intact. Scarring alopecia destroys follicles and replaces them with fibrosis; delayed diagnosis can lead to permanent loss. The basic evaluation includes pattern, tempo, symptoms, triggers, medication history, systemic symptoms, nutrition, endocrine signs, traction practices, family history, and scalp exam for scale, erythema, pustules, broken hairs, exclamation-point hairs, loss of follicular ostia, and perifollicular inflammation.

Alopecia type	Core clue	Mechanism	Evaluation	Management
Androgenetic alopecia	Gradual patterned thinning: bitemporal/vertex in men; central part widening in women.	Androgen-sensitive miniaturization of follicles in genetically predisposed patients.	Clinical pattern; consider hyperandrogenism if rapid onset, hirsutism, acne, irregular menses.	Topical minoxidil; oral options selected by clinician; antiandrogens in selected women; hair transplantation/camouflage.
Alopecia areata	Patchy, smooth, nonscarring hair loss; exclamation-point hairs; nail pitting can occur.	Autoimmune T-cell-mediated attack on hair bulb; associated with atopy and autoimmune disease.	Clinical; screen thyroid disease if symptoms/risk. Consider biopsy if atypical.	Intralesional corticosteroids for limited disease, topical agents; JAK inhibitors for severe disease in selected patients.
Telogen effluvium	Diffuse shedding 2-3 months after stressor: illness, surgery, childbirth, crash diet, medication, endocrine disease.	Synchronized shift of hairs into telogen phase.	Hair pull may be positive; evaluate ferritin/TSH/CBC based on context.	Remove trigger, reassurance, nutrition correction; regrowth takes months.
Anagen effluvium	Abrupt diffuse hair loss days-weeks after chemotherapy, toxins, radiation.	Injury to rapidly dividing matrix keratinocytes.	Medication/toxin timeline.	Usually reversible after trigger ends; scalp cooling in selected chemotherapy protocols.
Tinea capitis	Patchy alopecia with scale, broken hairs, black dots; kerion is boggy inflammatory plaque.	Dermatophyte infection of hair shaft/scalp; common in children.	KOH/culture; Wood lamp may help for some <i>Microsporum</i> species.	Oral antifungal therapy; topical shampoos reduce transmission but are not adequate alone.
Traction alopecia	Temporal/frontal hairline thinning related to tight hairstyles; fringe sign may remain.	Chronic mechanical tension; may become scarring if prolonged.	History and distribution.	Stop traction early; topical/intralesional anti-inflammatory therapy if inflamed; scarring areas may be irreversible.
Trichotillomania	Patchy hair loss with hairs of different lengths; bizarre geometric borders; usually no scale.	Compulsive hair pulling; psychiatric/behavioral association.	Clinical/dermoscopy; avoid punitive framing.	Behavioral therapy, psychiatric support; treat comorbid anxiety/OCD when present.
Lichen planopilaris / discoid lupus alopecia	Loss of follicular openings, perifollicular erythema/scale, dyspigmentation or atrophy.	Inflammatory scarring alopecia.	Prompt scalp biopsy at active border.	Dermatology referral; anti-inflammatory therapy to stop progression.

Alopecia red flags include scarring features, pain, pustules, boggy kerion, rapidly progressive diffuse loss with systemic illness, virilization, severe nutritional restriction, chemotherapy/toxin exposure, and signs of lupus or lichen planus. In children, tinea capitis should remain high on the differential for scaly alopecic patches because topical therapy alone is insufficient.

4. Nail Disorders

Nails are extensions of skin and often mirror systemic disease, inflammatory dermatoses, infection, trauma, and malignancy. The key exam distinction is whether the abnormality arises from nail plate, nail bed, nail matrix, periungual tissue, or distal digit. Nail matrix disease distorts the nail plate as it grows out; nail bed disease causes onycholysis, subungual debris, splinter hemorrhages, or pigmentation beneath the plate. A single changing pigmented nail streak requires a malignancy-focused approach.

Finding	Classic association	Key distinction	Clinical action
Onychomycosis	Thickened yellow brittle nail, subungual debris, onycholysis; toenails more common.	Can mimic psoriasis, trauma, lichen planus. Confirm with KOH, culture, PAS, or PCR before systemic therapy when possible.	Topical agents for mild disease; oral terbinafine commonly used for dermatophyte toenail disease after interaction/liver risk review.
Nail psoriasis	Pitting, oil-drop discoloration, onycholysis, subungual hyperkeratosis.	Often accompanies psoriasis or psoriatic arthritis; pitting can also occur in alopecia areata.	Treat psoriasis; consider joint symptoms and enthesitis screening.
Clubbing	Increased nail curvature and bulbous distal phalanx; loss of Lovibond angle.	Associated with lung cancer, chronic suppurative lung disease, cyanotic heart disease, IBD, liver disease; can be familial.	Evaluate for cardiopulmonary/GI/liver causes if new.
Koilonychia	Spoon-shaped concave nails.	Iron deficiency classically; also trauma/occupational exposure.	Check iron deficiency when clinically indicated.
Beau lines	Transverse grooves across nail plate.	Temporary arrest of nail growth after systemic illness, chemotherapy, severe stress, or local trauma.	Usually grows out; timeline can estimate prior illness.
Splinter hemorrhages	Longitudinal red-brown streaks under nail.	Trauma common; also endocarditis, vasculitis, psoriasis, antiphospholipid syndrome.	Interpret with fever, murmur, embolic signs, vasculitic symptoms.
Paronychia	Inflamed nail fold; acute painful bacterial versus chronic irritant/candida-associated inflammation.	Acute: abscess may need drainage. Chronic: wet work/irritant exposure; loss of cuticle.	Warm soaks, drainage if abscess, antibiotics when cellulitis; chronic requires irritant avoidance and anti-inflammatory/antifungal approach.
Longitudinal melanonychia	Brown-black longitudinal nail band.	Benign ethnic melanonychia often multiple/stable; melanoma more concerning when single digit, widening, irregular colors, Hutchinson sign, nail dystrophy.	Urgent dermatology referral/biopsy for suspicious features.

Subungual melanoma warning: Do not dismiss a new, widening, irregularly pigmented nail band as trauma unless it clearly grows out with the nail. Hutchinson sign, nail plate dystrophy, adult onset in a single digit, and extension onto proximal/lateral nail fold are high-risk features.

5. Benign, Vascular, and Premalignant Lesions

Benign lesions are common, but the clinician must avoid the cognitive trap of assuming all common lesions are harmless. A classic seborrheic keratosis can be observed, but a rapidly changing pigmented lesion that merely looks stuck-on should be examined carefully and biopsied if atypical. Benign labels should be reserved for lesions with stable history, classic morphology, and no red flags.

Lesion	Appearance	Pathophysiology / association	Management pearls
Seborrheic keratosis	Waxy, verrucous, stuck-on papule or plaque; tan, brown, black, or skin-colored.	Benign keratinocytic proliferation; inherited tendency; no malignant potential.	No treatment unless irritated/cosmetic. Biopsy if inflamed, bleeding, changing, or melanoma cannot be excluded. Sudden eruption of many lesions can rarely suggest Leser-Trelat sign.
Cherry angioma	Small bright red vascular papules on trunk/extremities; increase with age.	Benign capillary proliferation.	Reassure; remove for bleeding/cosmesis.
Dermatofibroma	Firm hyperpigmented papule/nodule, often legs; dimples inward with lateral compression.	Benign fibrohistiocytic reaction often after trauma/insect bite.	Observe if classic; biopsy if rapidly growing, ulcerated, atypical, or very large.
Epidermoid inclusion cyst	Mobile subcutaneous nodule with central punctum; may rupture and inflame.	Keratin-filled cyst from follicular infundibulum.	Do not call inflamed rupture infection automatically. Definitive treatment is excision of cyst wall when quiescent; drain if abscess.
Keloid / hypertrophic scar	Exuberant scar; keloid extends beyond wound boundaries while hypertrophic scar remains within original wound.	Excess collagen deposition after injury; genetic predisposition.	Intralesional steroids, silicone, pressure therapy, laser/surgery in selected cases; excision alone has high recurrence.
Actinic keratosis	Rough gritty scaly macule/papule on sun-exposed skin: face, ears, scalp, dorsal hands, forearms.	UV-induced atypical keratinocyte proliferation; premalignant SCC precursor.	Treat with cryotherapy for isolated lesions, field therapy such as 5-FU/imiquimod/photodynamic therapy for diffuse damage; biopsy tender, indurated, bleeding, or thick lesions.
Keratoacanthoma	Rapidly growing dome-shaped nodule with central keratin crater over weeks.	Often considered within SCC spectrum or SCC-like tumor.	Biopsy/excision usually required because it clinically resembles SCC despite possible spontaneous regression.

Actinic keratosis represents field cancerization: clinically visible lesions are the tip of a larger sun-damaged field containing subclinical atypia. This explains why patients develop new lesions even after individual lesions are treated. The most important referral triggers are pain, rapid growth, spontaneous bleeding, ulceration, induration, hyperkeratosis, immunosuppression, and failure to respond to standard therapy.

6. Nonmelanoma Skin Cancer: Basal Cell and Squamous Cell Carcinoma

Basal cell carcinoma (BCC) and cutaneous squamous cell carcinoma (SCC) are keratinocyte carcinomas strongly linked to ultraviolet radiation, fair skin, age, immunosuppression, arsenic exposure, radiation, HPV in selected sites, and chronic wounds or scars. BCC is the most common skin cancer and rarely metastasizes, but it causes local tissue destruction. SCC has greater metastatic potential, especially when high-risk features are present. Internal medicine clinicians should know the visual clues, biopsy threshold, and why transplant and immunosuppressed patients require aggressive surveillance.

Cancer	Typical clinical clues	Risk and behavior	Diagnosis and treatment concepts
Basal cell carcinoma	Pearly papule with telangiectasias; rolled border; central ulceration; pink nonhealing patch; scar-like morpheaform plaque; pigmented variant can mimic melanoma.	Slow-growing, locally destructive, rare metastasis. High-risk sites include central face, eyelids, nose, ears, lips.	Biopsy suspicious lesion. Treatment depends on risk: excision, Mohs for high-risk/anatomic sites, curettage/electrodesiccation for selected low-risk lesions, topical or radiation in selected cases.
Squamous cell carcinoma	Hyperkeratotic scaly papule/plaque/nodule; tender, indurated, ulcerated, or bleeding lesion; may arise from AK, chronic scar, burn, ulcer, radiation field, or immunosuppressed skin.	Higher metastatic risk than BCC. Risk increases with size, depth, poor differentiation, perineural invasion, ear/lip location, immunosuppression, recurrence.	Biopsy. Treat with excision or Mohs for high-risk lesions. Evaluate nodes when high-risk. Prevention and field therapy matter.
SCC in situ / Bowen disease	Well-demarcated erythematous scaly plaque, often persistent and solitary; can mimic eczema or psoriasis.	Intraepidermal SCC; can progress to invasive SCC.	Biopsy persistent plaque. Treat with excision, topical therapy, cryotherapy, photodynamic therapy depending on site/risk.
Merkel cell carcinoma	Rapidly growing painless red-purple firm nodule on sun-exposed skin in elderly or immunosuppressed patients.	Aggressive neuroendocrine carcinoma with high nodal/metastatic risk; associated with UV and Merkel cell polyomavirus.	Urgent biopsy/referral, staging, wide excision/radiation/systemic therapy as indicated.

Nonhealing rule: A lesion described as a persistent scab, sore, ulcer, or eczema patch that does not heal should be examined for BCC, SCC, SCC in situ, or melanoma. A steroid-responsive itch history does not rule out malignancy if the lesion persists or evolves.

7. Melanocytic Lesions and Melanoma

Melanoma is less common than BCC or SCC but carries the greatest mortality risk because it can metastasize early. The important teaching point is that melanoma is diagnosed by suspicion and biopsy, not by waiting for complete textbook appearance. Nodular melanoma may be symmetric and uniformly dark, amelanotic melanoma may be pink, and acral lentiginous melanoma may present on palms, soles, or under nails. The ABCDE rule is useful, but the ugly duckling sign - a lesion that looks different from the patient's other nevi - is often more sensitive in real patients.

Concept	High-yield detail
ABCDE	Asymmetry, Border irregularity, Color variegation, Diameter often >6 mm, Evolution. Evolution includes enlargement, bleeding, pain, itch, crusting, or color/shape change.
Ugly duckling	A mole that differs from the patient's signature pattern is suspicious even if it does not meet every ABCDE feature.
Biopsy approach	Preferred initial diagnostic test for a suspicious pigmented lesion is full-thickness excisional biopsy with narrow margins when feasible. Shave or partial biopsy can miss deepest invasion in some lesions.
Prognosis	Breslow thickness/depth of invasion is the most important local prognostic feature and guides staging and sentinel node decisions. Ulceration and mitotic rate also influence risk.
Major subtypes	Superficial spreading, nodular, lentigo maligna melanoma, acral lentiginous melanoma, and amelanotic melanoma. Acral lentiginous is especially important on palms/soles/nail units and in darker skin types.
Definitive management	Wide local excision margins are based on Breslow thickness. Sentinel lymph node biopsy is considered for appropriate tumor depths/risk features. Advanced disease may require immunotherapy/targeted therapy.

Benign nevi are typically symmetric, stable, uniform in pigment, and similar to the patient's other moles. Dysplastic nevi may be larger and irregular but are not melanoma by definition. However, a patient with many atypical nevi has higher melanoma risk and requires careful surveillance. A congenital nevus, especially large or giant, carries melanoma risk proportional to size and should be monitored. Blue nevi are usually benign dermal melanocytic lesions, but changing blue-black nodules still warrant evaluation.

Melanoma and pigmented lesion algorithm

Step 1: Ask whether the lesion is new or changing. Step 2: compare with surrounding lesions and patient photographs if available. Step 3: apply ABCDE and inspect special sites: scalp, ears, mucosa, palms, soles, interdigital spaces, and nails. Step 4: if suspicious, do not freeze, laser, or empirically treat; arrange full-thickness biopsy. Step 5: if pathology confirms melanoma, management is stage-based and requires definitive excision, nodal evaluation when indicated, and long-term surveillance.

8. Kaposi Sarcoma, Cutaneous T-Cell Lymphoma, and Cancer Mimics

Some malignant or systemic diseases initially look like routine dermatology. Kaposi sarcoma, cutaneous T-cell lymphoma, leukemia cutis, and metastatic disease can all present as persistent lesions that are misdiagnosed as bruises, eczema, psoriasis, infection, or benign nodules. Immunosuppression, HIV risk, systemic symptoms, lymphadenopathy, mucosal involvement, and treatment resistance should lower the threshold for biopsy.

Disease	Typical clues	Associations	Action
Kaposi sarcoma	Violaceous, red-brown, or purple macules, plaques, or nodules; may involve oral mucosa, legs, face, genitalia.	HHV-8; HIV/AIDS, transplant immunosuppression, classic Mediterranean/older men, endemic African disease.	HIV testing when appropriate, biopsy, immune restoration/ART in HIV-associated disease, oncology/dermatology therapy by extent.
Mycosis fungoides	Persistent patches/plaques in sun-protected bathing-trunk areas; can mimic eczema or psoriasis; may progress to tumors.	Cutaneous T-cell lymphoma; often indolent but diagnosis can be delayed.	Biopsy persistent or treatment-resistant dermatitis; multiple biopsies may be needed.
Sézary syndrome	Erythroderma, severe pruritus, lymphadenopathy, circulating malignant T cells.	Leukemic variant of CTCL.	Urgent dermatology/oncology evaluation with blood flow cytometry and staging.

Disease	Typical clues	Associations	Action
Leukemia cutis / metastasis	Firm papules, nodules, plaques, or violaceous lesions; variable appearance.	Known or occult malignancy, systemic symptoms, abnormal CBC.	Biopsy and systemic workup.

9. Integrated Diagnostic Algorithms

Presentation	Most important branch points	Likely diagnoses	Do-not-miss diagnosis
Diffuse hair shedding	Recent illness/childbirth/surgery/diet/drugs? Patterned thinning? Scalp inflammation?	Telogen effluvium, androgenetic alopecia, thyroid/iron disease, medication-related loss.	Scarring alopecia, systemic lupus, chemotherapy/toxin exposure.
Patchy hair loss	Scale/broken hairs? Smooth patch? Different hair lengths? Loss of follicular openings?	Alopecia areata, tinea capitis, trichotillomania, traction alopecia.	Kerion causing scarring, lupus/lichen planopilaris scarring alopecia.
White patches	Scale? Wood lamp accentuation? Sharp border? Distribution?	Vitiligo, tinea versicolor, post-inflammatory hypopigmentation, pityriasis alba.	Hypopigmented CTCL, leprosy in appropriate epidemiology.
Brown-black lesion	New/changing? ABCDE? Ugly duckling? Nail/acral location?	Nevus, seborrheic keratosis, lentigo, dermatofibroma.	Melanoma, pigmented BCC.
Rough scaly sun-exposed lesion	Gritty flat AK versus thick/tender/indurated lesion?	Actinic keratosis, SCC in situ, eczema, psoriasis.	Invasive SCC.
Pearly or nonhealing facial papule	Rolled border? telangiectasias? ulceration? bleeding?	Basal cell carcinoma, sebaceous hyperplasia, nevus.	Morpheaform/infiltrative BCC near eye/nose/ear.
Purple papules/plaques	Blanching? immunosuppression? oral lesions? leg edema?	Angioma, purpura, venous stasis, dermatofibroma.	Kaposi sarcoma, vasculitis, leukemia cutis.

10. Treatment and Referral Principles

A primary care or internal medicine clinician does not need to excise every lesion, but should recognize what not to treat empirically. Suspicious pigmented lesions should not be destroyed with cryotherapy because tissue diagnosis is lost. Nonhealing lesions should not receive repeated steroid courses without reassessment. Rapidly growing nodules, especially keratoacanthoma-like lesions, should be biopsied because SCC cannot be excluded clinically. Immunosuppressed patients should have a lower threshold for dermatology referral because SCC and HPV-driven lesions can be more numerous, aggressive, and recurrent.

Clinical situation	Preferred action
Classic isolated actinic keratosis	Cryotherapy or field-directed therapy depending on burden; reinforce sun protection and surveillance.
Actinic lesion is painful, indurated, ulcerated, bleeding, rapidly growing, or recurrent after therapy	Biopsy to evaluate SCC.
Classic seborrheic keratosis with stable history	Reassurance or cosmetic removal if desired.
Stuck-on lesion is inflamed, darkly pigmented, asymmetric, bleeding, or changing	Biopsy or dermatology referral to exclude melanoma or pigmented BCC.
Suspicious melanoma	Full-thickness excisional biopsy with narrow margins when feasible; urgent dermatology/surgery referral.
Single adult-onset longitudinal melanonychia with irregular band or Hutchinson sign	Urgent nail-unit evaluation and biopsy.
Scarring alopecia suspected	Dermatology referral for scalp biopsy at active border; treat early to prevent permanent loss.
Severe or extensive vitiligo/melasma/alopecia with psychosocial burden	Dermatology referral; discuss chronicity, relapse, photoprotection, and realistic endpoints.

11. High-Yield Differentiation Tables

Pair	How to separate quickly
Seborrheic keratosis vs melanoma	Seborrheic keratosis is stuck-on/waxy/verruccous and usually multiple/stable; melanoma evolves, may be asymmetric/variegated, and may not truly look attached to the surface. When uncertain, biopsy.
Actinic keratosis vs SCC	AK is rough/gritty and often thin; SCC is more likely tender, indurated, thick, nodular, ulcerated, bleeding, or rapidly enlarging.
BCC vs SCC	BCC is pearly, rolled, telangiectatic, ulcerated, and locally destructive; SCC is scaly/hyperkeratotic, tender, indurated, and has higher metastatic risk.
Vitiligo vs tinea versicolor	Vitiligo is depigmented, sharply demarcated, non-scaling, and Wood-lamp accentuated; tinea versicolor has fine scale and KOH positive yeast/hyphae.
Alopecia areata vs tinea capitis	Alopecia areata is smooth with exclamation hairs and no scale; tinea capitis has scale, broken hairs/black dots, lymphadenopathy, or kerion and requires oral antifungal therapy.
Onychomycosis vs nail psoriasis	Onychomycosis produces thick yellow brittle nail with subungual debris; psoriasis produces pitting, oil-drop changes, onycholysis, and may accompany plaques or psoriatic arthritis. Confirm fungus before oral antifungals when possible.
Kaposi sarcoma vs bruise/purpura	Kaposi lesions are persistent violaceous papules/plaques/nodules, often in immunosuppressed patients or with mucosal disease; bruises evolve color and resolve.

12. Review Cases

Case 1 - A changing mole

A 52-year-old with many nevi notices one lesion on the upper back has become darker and irregular over six months. It is 8 mm, asymmetric, and has brown-black color variation. The next step is not cryotherapy or observation; it is full-thickness excisional biopsy when feasible. Breslow thickness on pathology drives prognosis and definitive management.

Case 2 - White patches after summer

A young adult has multiple lightly hypopigmented patches on the upper trunk with fine scale. KOH shows short hyphae and spores. This is tinea versicolor. Pigment can take weeks to months to normalize even after antifungal therapy.

Case 3 - Patchy hair loss in a child

A child has scaly patches with broken hairs and posterior cervical lymphadenopathy. This is tinea capitis until proven otherwise. Oral antifungal therapy is required because topical therapy does not penetrate infected hair shafts adequately.

Case 4 - Rough lesion on ear

An older patient has a rough scaly tender papule on the helix that bleeds occasionally. Although actinic keratosis is possible, tenderness, bleeding, and a high-risk location should prompt biopsy for SCC.

Case 5 - Nail streak

A 60-year-old develops a new brown-black band on one thumbnail that is wider proximally and extends onto the nail fold. This is suspicious longitudinal melanonychia with Hutchinson sign and needs urgent nail-unit biopsy to rule out melanoma.

13. Graph-RAG Concept Map

Node / concept	Useful connected nodes	Relationship labels
Melanoma	ABCDE, ugly duckling sign, Breslow thickness, excisional biopsy, sentinel lymph node biopsy, acral lentiginous melanoma, subungual melanoma.	diagnosed_by; prognosticated_by; subtype_of; treated_by; red_flag_for.
Actinic keratosis	UV exposure, field cancerization, SCC, cryotherapy, 5-FU, photodynamic therapy.	precursor_to; caused_by; treated_by; risk_modified_by.
Seborrheic keratosis	Stuck-on lesion, Leser-Trelat sign, melanoma mimic.	benign_mimic_of; distinguished_from; associated_with.
Vitiligo	Autoimmune thyroid disease, Wood lamp, topical steroid, calcineurin inhibitor, JAK inhibitor, phototherapy.	associated_with; diagnosed_by; treated_by.
Alopecia areata	Autoimmune disease, exclamation-point hairs, nail pitting, intralesional steroid, JAK inhibitor.	associated_with; clinical_feature; treated_by.
Onychomycosis	Dermatophyte, KOH, culture, PAS, oral terbinafine, nail psoriasis mimic.	caused_by; diagnosed_by; treated_by; mimics.
Kaposi sarcoma	HHV-8, HIV/AIDS, transplant immunosuppression, violaceous plaques, ART.	caused_by; associated_with; treated_by; red_flag_for.
Mycosis fungoides	CTCL, chronic patches, bathing trunk distribution, skin biopsy, Sézary syndrome.	subtype_of; mimics; diagnosed_by; progresses_to.

References and source framework

Uploaded Medvale dermatology source: derm2.pdf, pages on viral infections, fungal infections, scabies, actinic keratosis, seborrheic keratosis, and keratoacanthoma.

American Academy of Dermatology. Melanoma clinical guideline highlights; biopsy and management principles for primary cutaneous melanoma.

American Academy of Dermatology. ABCDEs of melanoma and common skin cancer educational resources.

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Merck Manual Professional. Onychomycosis; Cutaneous T-cell lymphomas; dermatologic disorder references.

American Academy of Dermatology. Vitiligo: diagnosis and treatment overview.